

Affine-type pin v0.1 — $B_2^{(1)}$ vs $C_2^{(1)}$: substrate's natural affinization is $B_2^{(1)}$ (marks 1,1,2). Argument from keystone ($K=SO(5) \rightarrow \hat{so}(5)=B_2^{(1)}$); both give same finite Serre coefficients ($[2]_2=3$, $[3]_4=21$) so the finite algebra is the same; the affine choice is Langlands-dual labeling, decided by 'which loop algebra is the substrate's spatial-symmetry one.'

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0. The question

E5 (Elie) failed because the wrong affine type was built ($\tilde{A}_2$ marks 1,1,1 instead of correct $h=4$). Keeper corrected with $C_2^{(1)}$ (marks 1,2,1) from "highest root of C_2 is $\theta = 2\alpha_1 + \alpha_2$." My catch: $B_2^{(1)}$ (marks 1,1,2) is the alternate Langlands-dual affinization, naturally derived from B_2 's highest root $\alpha_1 + 2\alpha_2$. Both sum to $h=4$. Which is the substrate's?

1. Finite-algebra ambiguity (both work equally)

B_2 and C_2 are isomorphic as finite simple Lie algebras (both $= so(5) = sp(4)$). The choice between calling it " B_2 " or " C_2 " is purely a node-labeling convention. Elie's E0 + E9 verify the substrate Serre coefficients: - Short root: $[2]_{\{q^{d_{short}}\}} = [2]_2 = 3 = N_c$ (with $d_{short} = 1$) - Long root: $[3]_{\{q^{d_{long}}\}} = [3]_4 = 21 = N_c \cdot g$ (with $d_{long} = 2$)

BOTH B_2 and C_2 conventions reproduce these (since they're the same algebra). The finite Serre relations cannot distinguish B_2 from C_2 .

2. Affine ambiguity (Langlands-dual diagrams)

$B_2^{(1)}$ and $C_2^{(1)}$ are GENUINELY DIFFERENT affine algebras — Langlands dual:

	diagram	marks (a_0, a_1, a_2)	δ	affine attachment
$B_2^{(1)}$	$\alpha_0 - \alpha_1 \Rightarrow \alpha_2$	(1, 1, 2)	$\alpha_0 + \alpha_1 + 2\alpha_2$	SINGLE bond to LONG end (α_1)
$C_2^{(1)}$	$\alpha_0 \Rightarrow \alpha_1 \Leftarrow \alpha_2$	(1, 2, 1)	$\alpha_0 + 2\alpha_1 + \alpha_2$	DOUBLE bond to SHORT end (α_1)

Both have $h = \text{sum of marks} = 4$; both give $h^\vee = 3 = N_c$; both have the doubling in δ on the SHORT root. Different node labeling (B_2 : node 1 long, node 2 short; C_2 : node 1 short, node 2 long), different affine attachment

(B_2 : single bond to long; C_2 : double bond to short).

Their tube structures (the open #409 question) likely differ — this is part of why the count question is non-trivial.

3. The decisive argument (recommendation: $B_2^{(1)}$)

The keystone (#408, K-audit PASS) identifies $K = SO(5) \times SO(2)$ explicitly — the substrate's compact spatial symmetry is $SO(5)$, not $Sp(4)$. While $so(5) \cong sp(4)$ as Lie algebras, the geometric realization on D_{IV}^5 is $SO(5)$ -natural (the maximal compact factor of $SO_0(5,2)$ is $SO(5) \times SO(2)$, in the $SO(5)$ presentation, not the $Sp(4)$ presentation).

So the natural loop-algebra affinization is:

$$\hat{so}(5) = B_2^{(1)}, \text{ marks } (1, 1, 2)$$

$C_2^{(1)}$ would be the substrate's affine extension only if the geometric/Hall-algebra realization picks the $Sp(4)/C_2$ labeling — which it doesn't, per the keystone. The substrate's "affine direction" (the generation tower / imaginary-root extension) is naturally read off the $SO(5)$ loop algebra.

4. Cross-checks (consistency, not independent confirmation)

- Keeper's $C_2^{(1)}$ derivation is correct as math — $\theta(C_2) = 2\alpha_1 + \alpha_2$ in C-conventions IS the highest root. But this is the C-labeling. In the B-labeling (which the keystone uses), $\theta(B_2) = \alpha_1 + 2\alpha_2$.
- Both affines give $h=4$ — so the marks-sum criterion (Elie's E5 diagnostic for catching $\tilde{A}_2$) is satisfied by both. The pin is the labeling, not the sum.
- Both give $h^\vee = 3 = N_c$ — so the dual-Coxeter color identification (route II) is unchanged.
- The tube structure (#409) is affine-dependent — different tube counts for $B_2^{(1)}$ vs $C_2^{(1)}$ possible. This is part of why #409 needs the right affine pinned BEFORE recomputing (Elie's E5 failure mode).

5. Implications for the open work

- #409 tube count (route I, currently secondary per #414 reframe): if Elie reattempts on $B_2^{(1)}$, the simply-laced prior $\Sigma(\text{rank}-1) = V-2$ still gives ≤ 2 for $V=3$, but the non-simply-laced VALUATION of $B_2^{(1)}$ specifically (single bond to long + double pointing to short) may differ from $C_2^{(1)}$'s. Both still likely lean ≤ 3 , consistent with the gate disposition.
- L4 v0.2 explicit lepton mass ratios: Elie's bulk K-type radial tower for the spinor uses the affine δ -direction. For $B_2^{(1)}$, $\delta = \alpha_0 + \alpha_1 + 2\alpha_2$ with the doubling on α_2 (short, the spinor direction). This is the right structure for matter-tower computations.
- Generation tower / route II mechanism: the cross-cutting fermion tower (P1.6) lives in δ . $B_2^{(1)}$'s δ structure (doubling on short = spinor direction) is the natural seat for the spinor-tower investigation.

6. Honest scope + tier

RIGOROUS: $B_2^{(1)}$ and $C_2^{(1)}$ are distinct affine algebras (Langlands dual); both extend $B_2 \cong C_2 = so(5) \cong sp(4)$ finite; both sum to $h=4$; marks/ δ structures as stated; finite Serre coefficients don't distinguish them.

PIN RECOMMENDATION: $B_2^{(1)}$ (marks 1,1,2), based on the keystone identifying the substrate's spatial algebra as $SO(5)=B_2$ specifically. The substrate's natural affinization is the loop algebra of $SO(5)$, which is $B_2^{(1)}$.

HONEST CAVEAT: this is a CONVENTION decision driven by the keystone's labeling. $C_2^{(1)}$ would be the right choice if a geometric/Hall-algebra argument picked the $Sp(4)$ labeling instead. I don't see such an argument, but Keeper/Elie may.

Cal #27 / honesty: I'm NOT claiming this is forced by a deep argument — it's the natural choice given the keystone's $SO(5)$ labeling. If you think of the substrate as a $sp(4)$ loop, you'd pick $C_2^{(1)}$. The choice matters for the δ -tower structure and the tube count, so the team should ratify.

Routed: → Keeper: pin recommendation $B_2^{(1)}$ — concur or push back? Your $C_2^{(1)}$ derivation is correct as math; the difference is which labeling the substrate naturally uses. → Elie: once Keeper concurs/redirects, the bulk

K-type radial tower / L4 v0.2 work + any #409 recompute uses the pinned affine type. → me: continuing to P1.2 bulk-color push assuming $B_2^{(1)}$.

— Lyra, affine-type pin v0.1. RECOMMEND $B_2^{(1)}$ (marks 1,1,2; $\delta=\alpha_0+\alpha_1+2\alpha_2$) as the substrate's natural affinization, from the keystone $K=SO(5)\times SO(2) \rightarrow$ loop algebra $\hat{\mathfrak{so}}(5) = B_2^{(1)}$. Keeper-derived $C_2^{(1)}$ (marks 1,2,1) is the Langlands dual, same finite algebra under $Sp(4)$ labeling. Both sum to $h=4$; both give $h^\vee=N_c=3$; finite Serre coefficients don't distinguish. The pin decision matters for the δ -tower structure (doubling on α_2 short = spinor direction for $B_2^{(1)}$) and the tube count. Pending Keeper+Elie ratification.